

RATE & EXTENT – FACTORS AFFECTING RATE

KEY VOCABULARY

Concentration: more particles per volume → more collisions.

Temperature: more energy → faster, more successful collisions.

Surface area: smaller pieces → more area to collide with.

Pressure: (gases) higher pressure → particles closer together.

COMMON MISCONCEPTIONS

~~Heating makes more particles.~~

✓ Heating gives particles more energy, not more particles.

~~A bigger lump reacts faster.~~

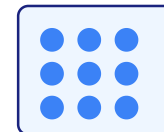
✓ Smaller pieces have more surface area → faster.

~~Concentration changes the amount of product.~~

✓ It changes the rate, not the final amount.

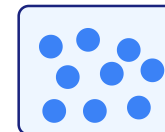
PARTICLES & COLLISIONS

SOLID



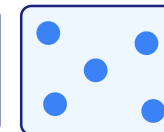
fixed, regular, touching

LIQUID



close, random, can flow

GAS



far apart, fast, random

More frequent, higher-energy **collisions** mean a faster rate.

1 LIST FACTORS

List **four factors** that affect the rate of reaction. [2 marks]

2 TEMPERATURE

Explain, using collision theory, why heating speeds up a reaction. [3 marks]

3 SURFACE AREA

Explain why powdered marble reacts faster than marble chips. [2 marks]

4 CONCENTRATION

Explain why increasing concentration increases the rate. [2 marks]

5 FILL THE GAPS

Complete the sentence using the word bank. [2 marks]

Raising temperature gives particles more ____, so collisions are more ____.

frequent

energy

fewer

slower

6 PRESSURE (GASES)

Explain why increasing pressure speeds up a reaction between gases. [2 marks]
